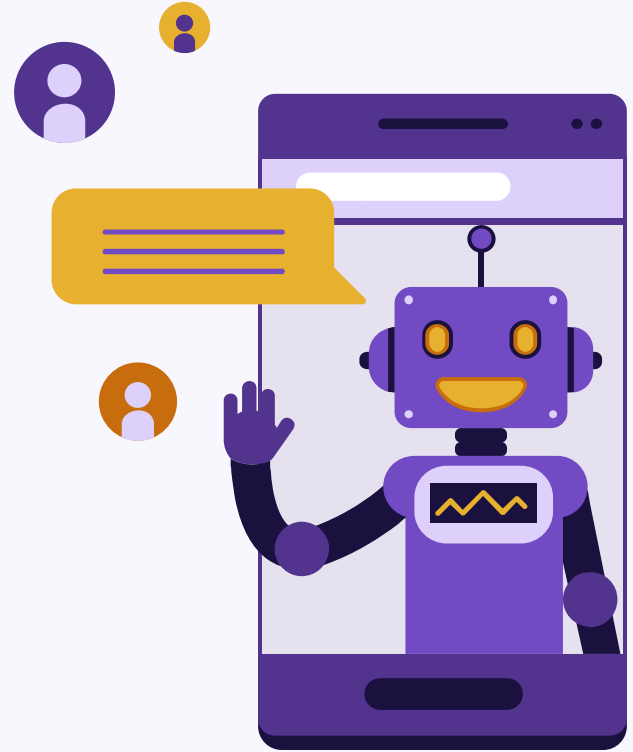


Characteristics of Artificial Intelligence in Scientific Writing

Jonathan, Leticia, Melle, and Wolter



Which is generated by an LLM?

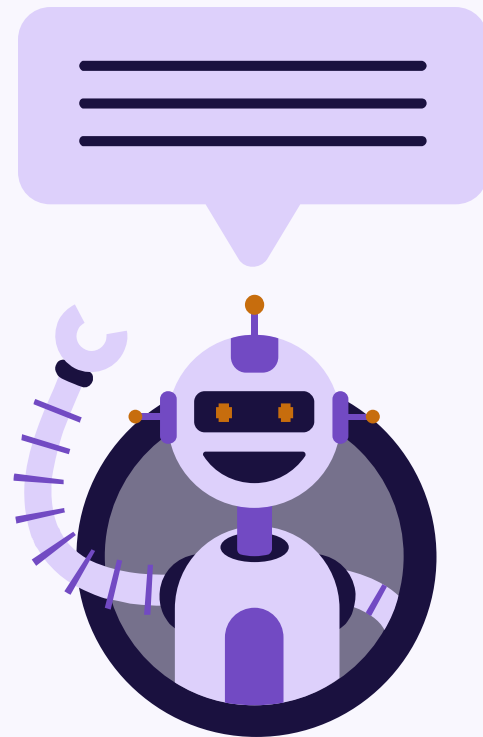
On the single-table complexity level with no join operations, SQL outperformed Polars for both aggregation tasks, obtaining significantly lower runtimes. These results were counterintuitive as Polars has the advantage of parallelization, which is often associated with faster runtimes. The slower runtimes observed for Polars are likely due to overhead related to the coordination of thread allocation and task scheduling required prior to running the query itself.

For straightforward, single-table aggregations that didn't require any joins, SQL surprisingly beat out Polars by delivering noticeably faster execution times. This outcome was unexpected because Polars utilizes parallel processing, a feature typically known for boosting performance. However, Polars likely suffered from initial lag caused by the system resources needed to manage threads and schedule tasks before actually running the code.

Introduction

- AI is now a co-author
- Writing made by LLMs goes unnoticed
 - **What characteristics of text are most indicative of it being AI-generated in scientific papers?**

“The rapid proliferation of sophisticated Large Language Models (LLMs) has blurred the lines between human and machine-generated text, posing significant challenges to academic integrity, content authenticity, and the combating of misinformation.” (Kingsley et al, 2025)



Dataset

Human Dataset

- 200 research articles from PubMed (published before 2020)
 - Stored as .txt files
 - Only abstracts and paragraphs
 - Remove XML tags

AI Dataset

- Feed the articles into the Gemini API with a prompt to rewrite in
 - Paragraph by paragraph

Real biomedical paper (from PubMed)

412 Inflammation and Carotid Atherosclerosis

Chronic low-grade inflammation has been increasingly recognized as a contributing factor in the progression of metabolic and cardiovascular disorders. Elevated circulating concentrations of interleukin-6 (IL-6) and C-reactive protein (CRP) have been associated with insulin resistance, endothelial dysfunction, and adverse cardiovascular outcomes in observational cohorts. Although the underlying mechanisms remain incompletely understood, both pathways converge on the regulation of hepatic acute-phase protein synthesis and the modulation of vascular permeability.

In the present study we examined the relationship between baseline plasma IL-6 concentrations and changes in carotid intima-media thickness (cIMT) over a 24-month follow-up period in a cohort of 412 adults aged 45 to 79 years without prior cardiovascular events. Anthropometric and biochemical measurements were obtained at study entry and at six-month intervals. cIMT was assessed by high-resolution B-mode ultrasonography by two trained sonographers blinded to participant status, with interobserver agreement evaluated using intraclass correlation coefficients.

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mass index, smoking status, and statin use, each one-standard-deviation increase in baseline IL-6 was associated with a 0.012 mm greater progression in mean cIMT (95% CI, 0.004 to 0.020; $P = 0.003$). The association persisted after additional adjustment for systolic blood pressure and LDL cholesterol. These findings support the hypothesis that systemic inflammation, as reflected by circulating IL-6, contributes to subclinical atherosclerotic progression independently of conventional cardio-



Gemini prompt: "Rewrite the paragraph below from the academic paper provided as cached context. Restructure the sentences and reorder the information; don't just swap synonyms. Don't introduce facts not in the original. Target ~N words."

Elevated circulating interleukin-6 (IL-6) and C-reactive protein (CRP) are linked to metabolic and cardiovascular disorders. These inflammatory markers, associated with insulin resistance and endothelial dysfunction, converge on hepatic protein synthesis and vascular permeability. While mechanisms are not fully understood, chronic low-grade inflammation is increasingly recognized as a progression factor.

This study investigated the link between initial plasma IL-6 levels and carotid intima-media thickness (cIMT) changes over 24 months. We analyzed 412 adults aged 45-79 without prior cardiovascular events. Measurements of anthropometrics and biochemistry were taken at baseline and every six months. Two blinded sonographers assessed cIMT using high-resolution ultrasonography, with interobserver agreement confirmed.

In this study, a one-standard-deviation increase in baseline IL-6 was linked to a 0.012 mm greater progression in mean cIMT over 24 months. This association remained significant even after accounting for systolic blood pressure and LDL cholesterol, in addition to age, sex, BMI, smoking, and statin use. These results suggest that systemic inflammation, indicated by IL-6 levels, independently drives subclinical atherosclerosis.

AI rewritten paper

412 Inflammation and Carotid Atherosclerosis

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Original

AI

Methodology Pipeline



Make Human Corpus

Download PubMed Research Papers using an **API**



Make AI/LLM Corpus

Feed these papers into the Gemini **API** and prompt it to rewrite them



Tokenize (**NLTK**) and extract text features (**SpaCy, POS tagging**)



Train a **logistic regression classifier**

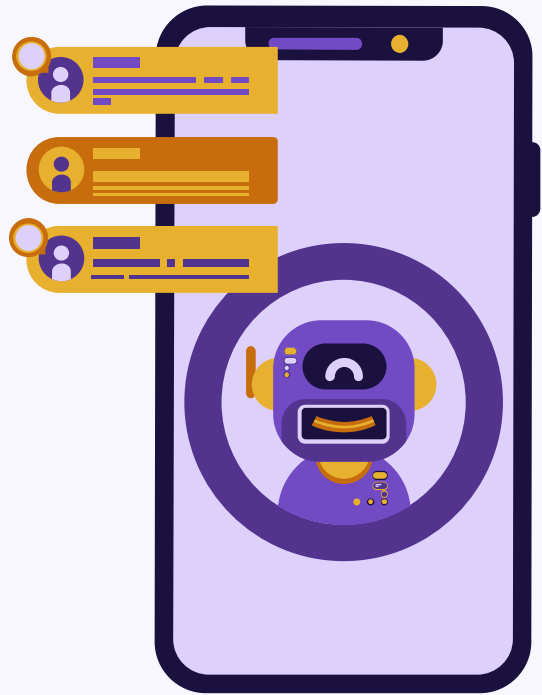
using the text features and labelled articles (human vs AI)



Obtain a classifier that, using the extracted features, can label unseen papers as being written by an LLM

Methodology - Features

- Average sentence length
- Average word length
- Type to token ratio
- The variation of sentence lengths (burstiness)
- Average parse tree depth
- Percentage of words that are associated with AI writing according to the literature
- Percentage of words that we found to be used more in our AI (train) dataset
- **Function word counts:** Pronouns, adpositions, determiners, conjunctions, and auxiliary verbs.



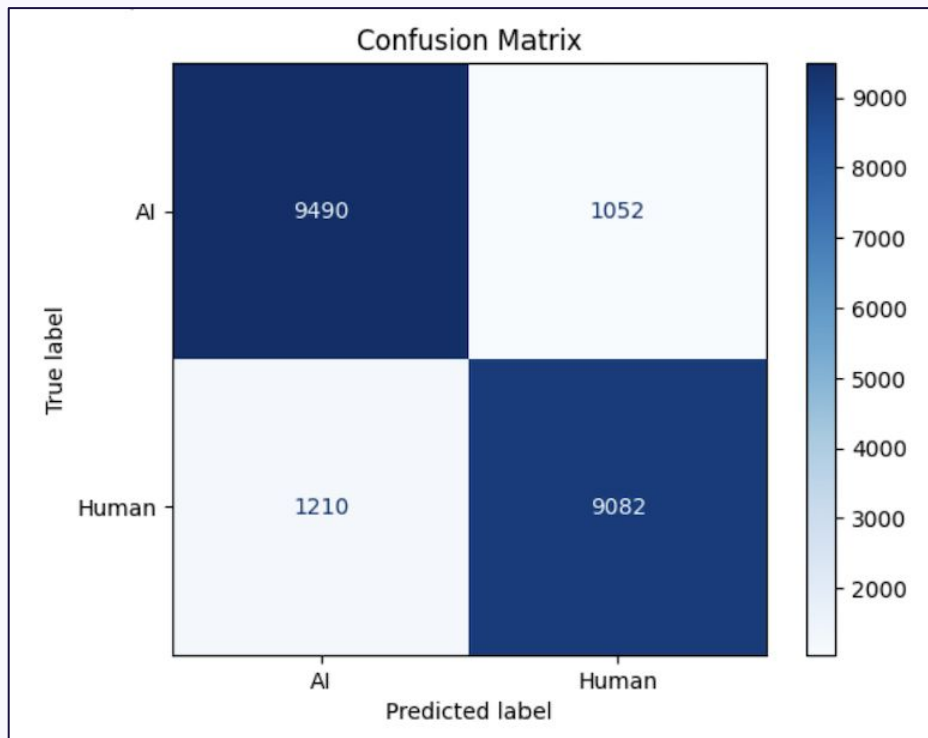
Results

- **Macro-F1: 0.891**
- **Accuracy: 0.891**
- Decently balanced model

Feature Importances (Higher is more human-like):

Pronoun rate	+0.7367
Burstiness	+0.6975
Auxiliary verb rate	+0.6664
Av Sentence Length	+0.6278
Av Parse Tree Depth	+0.4576
Conjunction rate	+0.3727
Preposition rate	+0.3300
Determiner rate	+0.2777
Av Word Length	+0.0280
Type/Token Ratio	-0.1489
Fixed AI Marker Rate	-0.1592
Learned AI Marker Rate	-3.8243

Intercept: [-0.71753075]



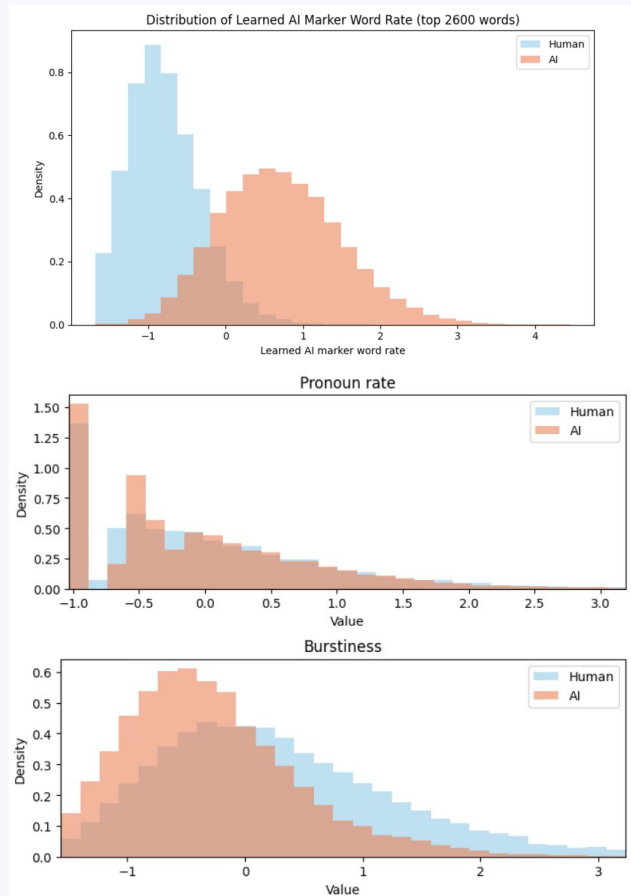
Feature Importance

- 3 most important features:
 - 1) Characteristic AI words (Learned AI Marker Rate)
 - 2) Pronoun rate
 - 3) Burstiness

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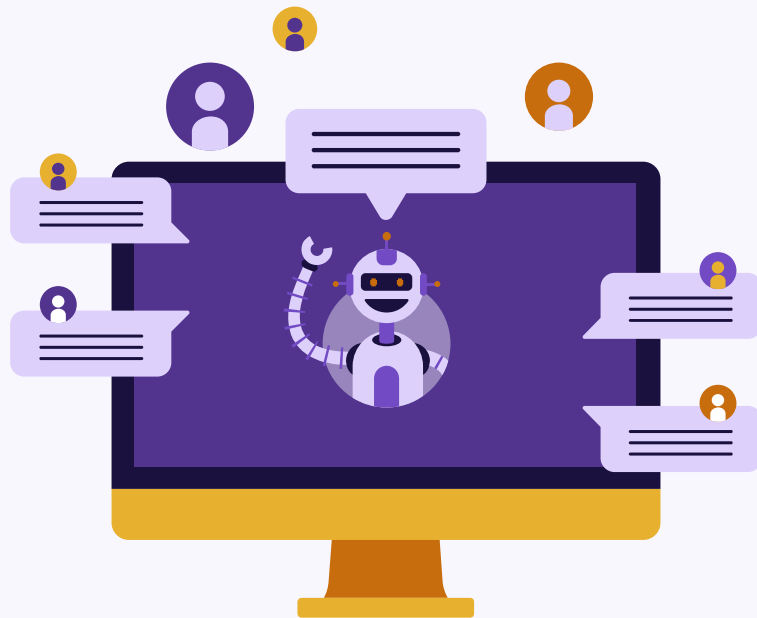
Limitations and Future Work

Limitations

- Relatively small dataset
- Solely PubMed articles
- Only Gemini was used to rewrite articles
 - Prompt used

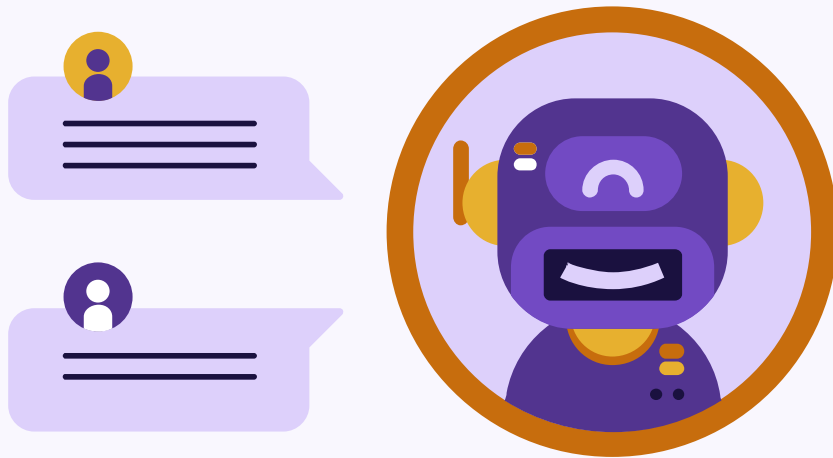
Future Work

- Test more models
- Include other languages



Conclusion

- What characteristics of text are most indicative of it being AI-generated in scientific papers?
 - Specific word use
 - Pronoun use
 - Varying sentence length
- **Macro-F1: 0.891**
- **Decent and balanced prediction**



References

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